

HexBoard Sequencer Quick Manual

HEXBOARD SEQUENCER MODE QUICK MANUAL

Overview

Sequencer mode is a separate mode from Keyboard mode. It is a single-track, 32-step sequencer. Each step can hold up to 4 notes, its own note length, its own velocity, its own probability, and can play through MIDI or the onboard synth.

Enter And Exit

- From the main menu, choose Sequencer.
- In the Sequencer menu, choose Keyboard to return to normal keyboard mode.

Main Controls

- Steps: the first 8 playable keys of the first 4 rows are steps 1-32.
- Play/Stop: the top-right key starts and stops playback on a short press.
- Play/Stop also deselects the current step if one is selected.
- Performance monitor: hold the top-right Play/Stop key for 2 seconds to show audio engine load, memory, storage, and MIDI receive stats; release it to close the monitor.
- Overview: the 9th key on the second row pages through an 8-step note view.
- Overview stays lit white while idle and becomes brighter white while the overview is open.
- Blue action key: the far-right key on the second row does undo on short press and clears the selected step on a 1-second hold.
- Tools: the last button on the third row opens a step-tools picker for the selected step.
- Tools stays lit white while idle and becomes brighter white while the tools picker is open.
- Encoder:
 - Turn while a step is selected to change that step's length quickly.
 - Press during file or folder naming to save the name.

Step Editing

- Press a step once to select it.
- Press the same step again to deselect it.
- With a step selected, play note keys on the lower part of the board to add or remove notes from that step.
- The edit screen shows a compact info line for the selected step: L ...% V ... P ...%, then the notes.

- Step notes are kept sorted from lowest to highest.
- Tap Preview can make a tapped step audition its stored notes when selected.

Step Tools Picker

- Press the last button on the third row with a step selected to open tools.
- The tools screen shows the selected step number, a compact L ...% V ... P ...% line, and the current notes.
- Current tool choices are: Len, Vel, Oct+, Oct-, Prob, Tie
- The tools screen also shows: Cancel/Finished
- Len opens exact length entry for the selected step.
- Oct+ transposes the selected step up 1 octave and stays in the tools screen.
- Oct- transposes the selected step down 1 octave and stays in the tools screen.
- Vel opens exact velocity entry for the selected step.
- Prob opens exact probability entry for the selected step.
- Tie is still a placeholder entry for now.
- Cancel/Finished closes the picker when you are done.

Step Velocity

- Velocity is stored per step.
- Range is 0-127.
- Default step velocity is 96.
- Use Vel from the step tools picker to edit it.
- Velocity entry uses the encoder.
- Turn the encoder to move through: 0, 5, 10, 15, ... 125, 127
- Press the encoder to save.
- Playback, tap preview, save, and load all use the stored step velocity.

Step Probability

- Probability is stored per step.
- Range is 0-100%.
- Default step probability is 100%.
- Use Prob from the step tools picker to edit it.
- Probability entry uses the encoder.
- Turn the encoder to move through: 0, 5, 10, 15, ... 100

- Press the encoder to save.
- Playback, save, and load all use the stored step probability.
- Tap Preview ignores probability and still auditions the step.

Note Length

- With a step selected, turn the encoder for quick length changes.
- Use Len from the step tools picker to open exact length entry.
- Available lengths are: 0%, 25%, 50%, 75%, 100%, 150%, 200%, 250%, 300%, 350%, 400%, 500%, 600%, 700%, 800%, 900%, 1000%

Exact length screen:

- press number keys to enter the value
- the first digit typed replaces the current value
- up to 4 digits are allowed
- <- deletes the previous digit and is on the third-row first key
- CANCEL exits without saving and is on the fifth-row first key
- press the encoder to save

Playback

- Tempo is the step clock.
- Each step is a 16th note at the current tempo.
- Hold Play/Stop for 2 seconds to show the performance monitor while held.
- Steps sets the active playback range from 1 to 32.
- Step lights beyond the active Steps count stay off, even if those steps still contain notes.
- Direction choices are: Forward, Backward, Ping-Pong, Random, Brownian, Drunk
- Play Type chooses: MIDI or OB Synth

Menu Items

- Save: saves to the current file.
- Save New: saves as a new file in the selected folder.
- Load: loads a saved sequence.
- Revert: reloads the last saved version of the current sequence.
- File Management: Rename File, Rename Folder, Create Folder, Delete File, Delete Folder, USB Backup
- Steps: active playback length

- Direction: playback order
- Tempo: BPM
- Tap Preview: audition step contents when a step is tapped
- Play Type: MIDI or onboard synth
- Synth Options: opens the shared synth settings page
- Update Firmware: reboots to the firmware update bootloader

Saving And Files

- Sequences are stored under /Sequences in onboard flash.
- On startup, the sequencer restores the last loaded sequence file when that remembered path is still valid.
- File Management > USB Backup starts a dedicated USB serial backup session for the / Sequences tree.
- The computer-side tools are: `scripts/hexboard_backup.py` for the command line and `scripts/hexboard_backup_gui.py` for a desktop GUI.
- The backup tools can work on the full /Sequences tree or on a selected file or folder.
- In the desktop GUI, you can multi-select remote items with Shift-click and delete them together.
- The GUI can rename the selected file or folder directly.
- GUI Upload Folder always merges into the target folder on the board. It does not delete the existing remote folder first.
- During GUI Upload Folder, any existing remote file conflict is shown one file at a time so you can overwrite, skip that file, or cancel the folder upload.
- If a remote file is selected, GUI Upload File warns before replacing that selected remote file and tells you to select a folder first if you want to copy into that folder instead.
- When multiple remote items are selected, single-target actions like Download, Upload, and Rename are disabled until only one item is selected again.
- The GUI remembers the last local folder used for Upload File and Upload Folder, and restores those locations after reopening the desktop tool.
- The top GUI status line shows active operations, including Refresh Remote.
- The Size Bytes column shows raw file sizes in bytes.
- While that session is active, Save, Revert, and file-browser actions are blocked until the session is stopped.
- File browsers show up to 8 file/folder rows at a time.
- ^ more ^ and v more v indicate earlier or later pages.

- Scrolling past the first or last real file row pages automatically.
- The menu title shows the current file name as: Seq-FILENAME
- If there are unsaved changes, it becomes: Seq-*FILENAME
- If no file is loaded, the title just says: Sequencer

Naming Mode

When naming a file or folder, the board shows this entry layout:

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z SPC - 1 2 3 <- CANCEL

- Press keys to enter characters.
- SPC adds a space.
- <- deletes the previous character.
- CANCEL exits naming without saving.
- Press the encoder to save the name.
- the physical CANCEL key is placed lower on the board to reduce accidental exits

Status And Display

- The selected step blinks.
- The overview screen shows 8 steps at a time.
- Empty steps are dark when idle.
- The performance monitor shows: AudioEng, Mem, FS, MIDI Q, Drop, and Late while the Play/ Stop key is held.
- Active sequencer overlays stay awake while you are using them.
- The top title shows the loaded file name and unsaved state.